Older than their fossils? What the Ordovician–Silurian record can test about the maxima that divide land-plant molecular clocks

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Abstract

Molecular clocks place the origin of crown land plants anywhere from about 515 to 450 million years ago (Ma), and of crown vascular plants from about 452 to 428 Ma. With the same sequence data, much of that spread follows from fossil calibrations, and above all from maximum ages: claims that a lineage would have left recoverable fossils earlier had it existed. We reassessed three such maxima, 515.5 and 459.1 Ma for crown land plants and 431.2 Ma for crown vascular plants, keeping fossil identity, age, sampling opportunity and the link from character to node separate. The 515.5 Ma maximum combines disputed fossil affinity with a qualitative preservation-control argument from older marine microfossil sequences; the material we assembled cannot quantify that argument. The 459.1 Ma value is the base of the Sandbian stage, which the oldest possible and the oldest definite Swedish trilete spores give alike. One Libyan basin supplies the only qualified quantitative comparison: 28 older well cuttings without trilete spores against 51 of 59 younger cuttings with them. If cuttings are independent and delivery was equal, it limits sustained, delivered trilete encounter in that basin, mostly through later-Ordovician samples, to about a tenth of the Llandovery level; treating cuttings from one well and formation as a block loosens the limit to about 0.4. It does not date a global absence. For crown vascular plants, 48 dated locality–age units give a 95% older bound of about 429 Ma on the start of their recoverable record if the rate of recorded, identifiable occurrences was constant or tracked collecting effort; shifting the stage boundaries within their uncertainties leaves it younger than 431.2 Ma. Coarser unit treatments move it to 432.4–434.05 Ma. For it to reach the 444–452 Ma of older clocks, identifiable bodies before the oldest find must have been recorded about 8 to 12 times less often than later, or 2.4 to 4.5 times under coarser units, which the record cannot test. Including Ordovician sporangial fragments from Oman moves the broad land-plant-body bound about 20 Myr older, to 455.2 Ma under constant intensity, and then no age assignment within the recorded ranges lets any tested history pass a shape diagnostic. These results make explicit which assumptions connect fossil records to maximum ages. They neither estimate a delay between plant origins and their bodies nor rank the competing clocks.

1. Why fossil maxima decide the age of land plants

When did plants colonise the land, and how fast did the vascular plants that came to dominate it diversify? Molecular clocks give strikingly different answers (Fig. 1). Morris et al. (2018b) placed crown Embryophyta (the last common ancestor of living land plants and all its descendants) at 515–474 Ma across seven topologies, and Harris et al. (2022) at 515–493 Ma. Hooper et al. (2026), using the same molecular dataset as Harris et al. but recalibrated fossils, obtained 457–450 Ma: a crown-group mean 55 Myr younger. For crown Tracheophyta (living vascular plants) the estimates are 451–419 Ma (Morris et al.), 452–447 Ma (Harris et al.) and 432–428 Ma (Hooper et al.). Su et al. (2021) obtained still older crown-embryophyte ages with a Proterozoic maximum. These are disagreeing results, not one uncertain estimate.

The difference matters biologically. If crown land plants are Cambrian, they existed for about 80 Myr before the oldest branched land-plant sporophyte, the Sheinwoodian *Cooksonia barrandei*, about 432 Ma (Libertín et al. 2018), and some 55 Myr before the oldest accepted sporangial fragments, from the Katian of Oman (Wellman, Osterloff, and Mohiuddin 2003). If they are Late Ordovician, those fragments and the oldest branched sporophytes sit close to the root of the living radiation. The two readings imply different histories of terrestrial productivity, weathering and the assembly of the plant body plan (Kenrick et al. 2012; Strother and Foster 2021; Donoghue et al. 2021).

Fossils enter clocks as minima and maxima. A minimum rests on a dated fossil whose placement can be argued. A maximum rests on what was not found: older rocks that should have yielded the diagnostic fossil if the lineage had existed. Hooper et al.'s recalibration shows how much rides on that argument. Morris et al. set the crown land-plant maximum at 515.5 Ma, below the oldest possible cryptospores, and argued that older sequences with marine microfossils but no land-plant spores record a real absence. Hooper et al. set it at 459.1 Ma from the oldest possible trilete spores, and set 431.2 Ma for crown vascular plants. The use of absence-based maxima is itself contested (Hedges et al. 2018; Morris et al. 2018a; Su et al. 2021).

A maximum makes two claims that are easily run together (Fig. 2). The first concerns recovery: that sampled rocks of the right age, facies and preparation would have yielded the fossil if its producers were present at some stated abundance. The second concerns biology: that the producing lineage made the diagnostic spore

or body from its origin, so the first recoverable fossil can stand in for the node. Marshall (2019) separates the same two gaps: from divergence to the first diagnosable character, and from that character to its first fossil. Early land plants strain both. Dispersed land-plant spores appear long before branched land-plant bodies (Wellman and Gray 2000), with fragmentary sporangia between them. Terrestrial deposits are scarce before the latest Silurian, at least in western Europe (Kenrick et al. 2012). The oldest parent plants known in detail are minute, and bryophyte-grade plants lack lignin and preserve poorly (Dianne Edwards et al. 2014; Morris et al. 2018b).

We therefore asked what each of the three maxima would need from the sampled record, and then tested what the record allows. The work had three parts:

1. **An audit** of the observations behind each maximum, keeping the specimen, its identification, the sampling evidence, the persistence of the diagnostic trait and the node it bears on as separate links.
2. **An absence test** wherever an older-empty, younger-positive spore comparison could be built from documented samples.
3. **A positive-record analysis** of dated body fossils, which bounds how far before the oldest find the recoverable record of a fossil category could have begun, under declared histories of the rate of recorded, identifiable occurrences.

The third part needs no inventory of searched but empty rock; it substitutes an assumption about that rate before the first find, and we vary the assumption explicitly. The rate combines how common and widespread the producers were, whether they had recognisable characters, delivery and preservation, and collection and reporting; these are not separated here.

2. Evidence, its geography and its independence

Three bodies of evidence carry the tests (Fig. 3).

Murzuq Basin cuttings. Abuhmida (2013) reported palynology from Ordovician to lower Silurian wells in southwest Libya. We transcribed all 164 appendix samples and their chart and prose reports, and used the chart-coded cuttings of five wells as the primary frame. Three wells supply 28 cuttings older than the Llandovery, none reporting trilete spores, and all five supply 59 Llandovery reference cuttings, 51 of which report them. Northern Africa lay near the Ordovician South Pole, within the area of Hirnantian glaciation (Cocks and Torsvik 2021; Scotese et al. 2021), so this frame samples one high-latitude basin.

QSIM-801. Vecoli et al. (2017) tabulate 99 cryptospore occurrences in 17 of 20 samples from a Saudi Arabian borehole assigned to the late Dapingian–early Darriwilian, and report no trilete spores in the studied sequence. The material and basin are independent of Murzuq.

Dated body fossils. We assembled the accepted land-plant body occurrences from 453 to 393 Ma that could be dated and linked to a locality. Land-plant bodies were declared as any accepted macro- or mesofossil, sterile axes included. The inventory as first built took curated records only from branched-axis, tracheid and crown-member channels, and so omitted the spore-filled sporangial fragments from the Ghaba-1 borehole, Oman (Wellman, Osterloff, and Mohiuddin 2003), which the declared category admits. We include them as one locality–age unit, Katian on the later review's correlation (Caradoc in the original paper), and report the former inventory as an exclusion sensitivity (§3.3). Sources were the Paleobiology Database (PBDB) names, classified by AI with cited sources and audited, together with our curated set of the oldest records and their competing readings. The audit also covered a separate screen of all 556 older non-spore PBDB rows. The counting unit is a locality–age proxy, so several collections, taxa or papers from one horizon count once. Without the Oman unit, the main reading yields 77 land-plant body units, 69 of polysporangiophytes (branched sporophytes with several sporangia) and 50 of crown vascular-plant members (the lycophyte line or euphyllophytes); 75, 67 and 48 are certainly older than the end of the Emsian (393.47 Ma). The Oman unit adds one to the land-plant count only (76 certain). A polysporangiophyte unit needs branching or crown-member characters; tracheids alone do not admit one, a rule that an earlier build broke for three records (Supplement S5.1).

The body record is geographically concentrated where it matters most. Of the 48 crown vascular-plant units, only 16 have age support older than the Silurian–Devonian boundary, and those come from seven regions.

The two oldest-possible clusters are Yea, Victoria, two horizons whose age is disputed between Gorstian and Early Devonian (Earp 2024), and the Ludfordian of Bathurst Island, Arctic Canada (Kotyk et al. 2002). Silurian land-plant bodies more generally come mainly from Laurentia (11 of 34 wholly Silurian units) and peri-Gondwanan Europe (8). Laurentia supplies 29 of the 78 land-plant units, most of them Devonian.

Three comparisons are independent of one another in material: Murzuq, QSIM and the body record. Within each, independence is limited. Murzuq's 87 cuttings come from five wells, and its older cuttings from three. Body units share regions and collectors, and their proxy for a horizon is not a verified bed. None is independent of the clocks in the way a validation would need. The oldest *Cooksonia* and Ludlow *Zosterophyllum* are themselves calibrations, and the oldest trilete spores set a maximum. Agreement between a body-record bound and a clock therefore tests whether an absence-based maximum is reasonable; it does not validate a clock posterior or rank one clock above another.

Table 1 sets out the records on which the conclusions turn: what each shows directly, how we use it, and what a crown claim would additionally need.

Table 1. Pivotal records.

Record	Observed or documented	Age support	Our use	A crown claim would also need
Ghaba-1 fragments, Oman (Wellman, Osterloff, and Mohiuddin 2003)	Sporangial fragments with in situ cryptospores, some with a sporangial covering; no axes or conducting tissue	Katian on later correlation (Caradoc in the original)	Land-plant body (B0), one unit	Characters placing the producer in a crown group; not B1 or B2
Borenshult-1 triletes, Sweden (Rubinstein and Vajda 2019)	Possible trilete at 48.2 m; definite triletes from 43.1 m; the figures disagree at 41.1 m	Sandbian; the oldest definite level lies below the 454.41 Ma bentonite	The 459.1 Ma maximum's basis; a weak absence screen	An argument placing the producer in the crown; a trilete mark alone does not
Cooksonia barrandei, Czech Republic (Libertín et al. 2018)	Branched fertile sporophyte with terminal sporangia and in situ trilete spores; no tracheids reported	Sheinwoodian, horizon inferred from slab fauna and correlation	Oldest polysporangiophyte (B1)	Placement on the lycophyte or euphyllophyte side; tracheids alone would not suffice
Irish Homerian axes (D. Edwards and Feehan 1980)	Fragmentary sporangia on short axes, one branched; no tracheids seen	Late Homerian graptolite zone	Hooper et al.'s 431.2 Ma anchor	As for *C. barrandei*
Yea *Baragwanathia*, Victoria	Lycopsid characters (microphylls, sporangia)	Gorstian on graptolites versus Early Devonian on brachiopods and mapping (Earp 2024); full range carried	Influential crown vascular-plant (B2) unit	Matching the competing dates to the exact horizons

Bathurst Island zosterophylls, Arctic Canada (Kotyk et al. 2002)	Fertile zosterophylls in spikes and clusters	Late Ludlow on conodonts, graptolites and brachiopods; possibly Pridoli	Influential B2 unit (lycophyte line, within crown Tracheophyta)	None for crown Tracheophyta; not crown Lycophyta

3. Results

3.1 Crown land plants before 515.5 Ma: affinity and a qualitative preservation-control argument

The 515.5 Ma maximum of Morris et al. (2018b) has two parts. Its placement is set below the oldest possible total-group records: cryptospore-like packets and polyads from the Cambrian Stage 4 Rome Formation, so the base of the *Olenellus* Biozone, about 515.5 Ma. Its absence argument is that land-plant spores share the preservation potential of the marine algal cysts with which they usually occur, and that older sequences with such cysts but no land-plant spores therefore record a real absence (Morris et al. 2018b, 2018a). Our earlier drafts described only the first part.

On affinity, none of the three sources we examined accepts the Stage 4 material as land-plant spores. Strother (2016) placed the Tennessee taxa in Charophyta, Taylor and Strother (2024) read the packets as pre-embryophytic, and Wang, Xu, and Yin (2022) favoured ancestral aeroterrestrial algae while calling an embryophyte affinity unlikely. That the material is younger than the maximum (514.5–506.5 Ma) is by design and is not a criticism.

On absence, the control shows that sporopollenin-walled microfossils survived and were recovered. It does not by itself show that terrestrial spores were delivered to those marine basins at comparable rates, that the sampled basins drew on the land areas where early plants grew, that preparation and search would have found small cryptospores, or that the earliest producers made recognisable dispersed spores. No source we assessed supplies an enumerated, comparably searched pre-515.5 Ma sequence with sample-level counts from which a non-detection probability could be calculated. The argument is therefore qualitative: we can state which links it leaves unquantified, but not attach a calibrated absence probability to it.

3.2 Crown land plants before 459.1 Ma: a local test of sustained encounter

The 459.1 Ma maximum of Hooper et al. (2026) rests on the oldest possible trilete spores, from the Borenshult-1 core in southern Sweden (Rubinstein and Vajda 2019). Their Supplementary Table 4 keys it to a queried *Chelinospora?* specimen at 48.2 m and takes the number from the stage, not the level: the GTS2012 base of the Sandbian plus its uncertainty, 458.4 + 0.7 = 459.1 Ma. The describing authors plot that specimen as a possible trilete only. They plot definite triletes from 43.1 m upwards, still in the Sandbian and below the 454.41 Ma Kinnekulle K-bentonite (Supplement S2.4). The stage-base rule gives the same maximum from either level, so the value does not rest on the doubtful specimen. Accepting the definite triletes could move a minimum, not this maximum, and only with a further argument that their producers were crown members, which a trilete mark alone does not supply. As an absence test the core is weak: the three Darriwilian samples below its oldest definite trilete cannot exclude reference-level intensity under the screen we applied, under any of the three readings of its figures (). It remains direct, dated character evidence.

Hooper et al.'s ancestral-state reconstructions make a proximal trilete aperture modal at the embryophyte crown (aperture and dispersal unit were reconstructed separately; Supplement S6), so a crown older than 459.1 Ma would imply undetected trilete producers. Murzuq is the only frame we found in which that implication can be confronted with documented older and younger samples. Its source describes the microscopy protocol and states the absences explicitly, so the zeros are observations, not database gaps (Abuhmida 2013). Their timing matters, though: of the 28 older cuttings, 10 are Darriwilian, broadly dated and from two wells, and 18 are Katian or Hirnantian, wholly younger than 459.1 Ma.

We modelled each cutting's chance of reporting triletes as an encounter process sharing a well effect, with older encounter intensity expressed as a fraction f of the Llandovery reference. An assumed delivery-and-recovery ratio ρ covers differences in transport, preservation and preparation between older and younger material. We call a procedure *qualified* when its simulated coverage reached 0.93 on a declared grid of generating scenarios

(Supplement S2); this is a check of the method on that grid, not a verification of the historical sampling process.

Table 2. What the Murzuq comparison allows: model-conditional 95% compatibility limits, assessed on the declared simulation grid, on older trilete-producer intensity as a fraction f of the Llandovery reference, primary stage allocation. Rows 1–2 constrain intensity sustained from 459.1 Ma to the reference; rows 3–4 constrain an interval of fixed intensity beginning at 459.1 Ma and extending further back.

Question asked of the 28 older cuttings	Delivery equal ($\rho = 1$)	Delivery halved	Delivery quartered	Delivery a tenth
Largest constant producer intensity, 459.1 Ma to the Llandovery	0.091	0.182	0.364	0.910
Largest intensity, exposure-weighted history class	0.184	0.258	0.421	0.921
Oldest end of a reference-intensity interval beginning at 459.1 Ma	462.6 Ma	464.1 Ma	467.1 Ma	no limit
Oldest end of an interval at 0.1 or 0.01 of reference intensity	no limit	no limit	no limit	no limit

The result is clear in form and conditional in substance (Table 2). What the zeros constrain is delivered, recovered encounter, sustained over the later Ordovician in one basin, since most exposure comes from the 18 younger cuttings. Under the constant history the product ρf stays at 0.091 across delivery scenarios; under the exposure-weighted history it runs from 0.184 at equal delivery to 0.092 at a tenth. If older material delivered and preserved spores as well as the Llandovery reference, trilete producers at constant intensity from 459.1 Ma must have been below a tenth of their Llandovery intensity, and below about 0.18 of it under the exposure-weighted history. This bounds sustained intensity in one basin, not the intensity at every time in the Late Ordovician. If delivery was a tenth as good, producers at nearly the reference intensity are allowed under either history. Intervals of producers at a tenth or a hundredth of reference intensity are unbounded under every delivery scenario. Under the model, the record therefore excludes abundant, well-delivered trilete producers sustained through the later Ordovician in this basin, not sparse or poorly delivered ones, and it cannot tell those two apart. Nor can it tell either from a lineage that lived outside the basin's catchment: a local absence of delivered spores is not a global absence of producers.

The verdict also moves with the auxiliary assumptions. Removing one well at a time spans exposure-weighted limits of 0.185–0.365. The pre-maximum window rests on the ten Darriwilian cuttings, and how their broad stage ages are turned into exposure decides it: shared midpoint ages give the 459.1–462.6 Ma window no exposure at all. With no contaminant transfer to older cuttings, a reduction of more than about 4.4% in the indigenous share of reference intensity crosses the short window's comparison threshold. At that window's simulated boundary, 20 of 1,000 screen-passing simulations missed the truth, and those 20 were among only 60 that made a claim. That conditional miss rate is reported for transparency; it is not by itself a failure of the unconditional procedure, but it shows that a finite boundary claim from this frame should not be read as an unrestricted 95% guarantee.

Dependence among cuttings matters as much. Cuttings from one well share a formation and a stage assignment, and a shared well effect need not absorb that. Fitting a separate rate to each well barely moves the constant-history limit (0.0995 against 0.0910). Down-weighting each cutting by the size of its well–formation or well–stage block, so that the older record counts as about six blocks rather than 28 cuttings, loosens it to 0.393 and 0.419. These are retrospective composite-likelihood sensitivities outside the qualified procedure, but they show that "a tenth" rests on treating cuttings from one well and formation as independent observations (Supplement S2.5, Fig. S2). The comparison is real and useful, but it is one basin at a glacial pole, it depends on preservation, delivery and dependence, and its reference positives are all cuttings.

QSIM-801 points the same way qualitatively: its authors report no triletes in a cryptospore-rich Middle Ordovician sequence. A numerical version does not reach the same standard. Treating each of the 99 occurrences as an independent unit on which a trilete would have been detected, a trilete-to-cryptospore

ratio that stayed above 0.031 in every sample is incompatible with the record at equal effective recovery, and one above 0.307 at a tenth of it (as shares of all spores, 0.030 and 0.235). This bounds the minimum ratio across samples, not an average or the pooled assemblage. The source does not document the search protocol needed to make those units independent and target-detectable, so we report this as an exploratory, model-conditional calculation, not a second qualified test (Supplement S3).

3.3 Crown vascular plants: what the body record allows

No assessed source supplies a roster of searched Silurian beds with comparable effort for the 431.2 Ma maximum. The Irish late-Wenlock sections report fossil-free mudstone qualitatively, without an enumerated roster (D. Edwards and Feehan 1980), and the historical Barrandian axes are selected specimens, not search trials. We therefore turned to the positive record.

The fossil basis. Hooper et al. anchor the maximum on the Irish Homerian *Cooksonia*, taking the GTS2012 Homerian base plus its uncertainty, 430.5 + 0.7 = 431.2 Ma. Their calibration table does not discuss the older *C. barrandei* (Table 1). On the same characters, a branched sporophyte with terminal sporangia and no demonstrated tracheids, it qualifies at least as well, and its Sheinwoodian horizon, inferred from slab fauna and correlation, would move the anchor by up to about 3 Myr (Sheinwoodian base 432.9 Ma). An older total-group fossil does not contradict a younger crown maximum. The point is narrower: on the authors' own rule, the absence on which the maximum rests begins earlier than stated.

Method. Let $w(t)$ be a fixed relative rate of recorded, identifiable occurrences of a fossil category: it combines how common and widespread the producers were, whether they bore recognisable characters, delivery and preservation, and collection and reporting, none of which is separated here. For n independent units older than a fixed younger cutoff c, the one-sided 95% older bound T on the start of the recoverable record solves $U(T) = U(A)\,0.05^{-1/n}$, where A is the oldest unit's age and $U(a) = \int_c^a w(s)\,ds$. With constant w this is the classical confidence interval on a stratigraphic range (Strauss and Sadler 1989; Marshall 1990), and the absolute rate cancels. Interval-dated units enter conservatively: the oldest possible age and the smallest number certainly older than the cutoff. The bound is a confidence statement under a fixed observation model and the adopted numerical timescale, not a posterior probability for the crown. We report six histories: constant intensity; intensity proportional to PBDB land and marginal-marine collections per Myr, with and without collections that match the plant-filtered extract; and intensity declining exponentially into the past on 25 and 10 Myr timescales, the last also combined with plant-excluded collection density (Supplement S5).

Crown vascular plants. For the 48 crown-member units the bounds are close together under the histories tied to the observed record: 428.8 Ma for constant intensity, 428.9 Ma for collection density and 428.5 Ma with plant-matched collections removed (Table 3). A 25 Myr decline into the past gives 431.6 Ma, and a 10 Myr decline gives no bound within the 469.4 Ma limit of the effort history. The unit definition matters as much. Under constant intensity, collapsing units to region–stage cells (19 units) gives 432.4 Ma and the curated oldest-record set alone (15 units) gives 434.05 Ma, both older than 431.2 Ma. These are not estimates of an effective sample size, and the curated set also changes what was searched, but they show that plausible ways of organising the same records move the bound across the maximum. Other declared sensitivities stay close to the primary value: individual collections 428.0 Ma, wide-age records restored 428.1 Ma, each region left out in turn 427.2–430.5 Ma, and a window ending at the Silurian–Devonian boundary (13 units) 428.5 Ma.

Two further checks bear on the comparison with 431.2 Ma. Shifting every stage boundary together by its quoted uncertainty, or alternately in opposite directions, moves the constant-intensity bound between 427.31 and 430.42 Ma, by at most 1.58 Myr. It stays younger than the maximum whether 431.2 Ma is held fixed, moved with the shifted Homerian–Sheinwoodian interval, or recomputed by Hooper et al.'s own stage-base-plus-uncertainty rule; under the 25 Myr decline it crosses. These are deterministic scenarios, not a probability interval. Removing the three Yea units changes the constant bound by −1.67 Myr, and removing the five Bathurst Island units by 0.26 Myr (positive is older), because the count falls while Yea stays oldest. Removing *C. barrandei* changes the polysporangiophyte bound by −2.38 Myr (Supplement S5.3).

Table 3. Conservative 95% older bounds (Ma) on the start of the recoverable record of each body category, main reading, conditional on each recording history, the unit rule and independent units. The land-plant

body column includes the Katian Oman sporangial fragments; the next column omits them. † One illustrative scenario chosen after the main analysis; not fitted, not a threshold and not a nominal 95% bound. "None": no bound within the 469.4 Ma limit of the effort history.

Recording history or unit rule	Land-plant body	Land-plant body, without Oman	Polysporangiophyte	Crown vascular plant
Constant	455.2	434.5	434.7	428.8
Proportional to PBDB collection density	456.6	436.4	436.8	428.9
Collection density, plant-matched collections removed	456.1	435.7	436.1	428.5
Declining into the past, 25 Myr timescale	465.2	437.2	437.7	431.6
Declining into the past, 10 Myr timescale	none	none	none	none
Constant, region–stage units	—	—	—	432.4
Constant, curated oldest-record set only	—	—	—	434.05
Constant, earlier intensity reduced tenfold before 426.7 Ma†	—	446.7	448.4	448.1
Constant, earlier intensity reduced a hundredfold before 426.7 Ma†	—	none	none	none

Land-plant bodies and the oldest admitted unit. Including the Ghaba-1 fragments as one unit, with oldest possible age 452.8 Ma, gives land-plant body bounds of 455.2–456.6 Ma under constant and effort-tracking intensity and 465.2 Ma under the 25 Myr decline. Without them, the oldest unit is the Sheinwoodian *C. barrandei* and the bounds are 434.5–437.2 Ma (Fig. 4, top). One unit moves the bound by about 20 Myr. With it, every declared history fits the sequence poorly (constant intensity: nominal $p = 5.5 \times 10^{-12}$ at midpoint ages), because the Katian unit is followed by more than 12 Myr without another, and no assignment of ages within the recorded ranges rescues any of them. These bounds are therefore reported as sensitivity arithmetic, not as a date at which land-plant bodies became recoverable. Comparing them with the 458.2 Ma minimum for land-plant cryptospores does not estimate a delay between origin and recoverable bodies in either direction: that value is a fossil minimum, not an origin, and the bounds are model-conditioned endpoints. The useful lesson concerns the crown vascular-plant bound too, which rests in the same way on its oldest units, Yea and Bathurst Island.

What the record can and cannot test. The shape diagnostics compare the observed age sequence with each history, using asymptotic tests on ages allocated within their reported ranges and then treated as exact. A steep (10 Myr) decline fits the crown-member sequence poorly under every allocation drawn: p of about 10^{-11} to 10^{-6} across the 5th–95th percentiles of independent allocations and 3×10^{-13} at midpoint ages. Constant intensity looks adequate only under independent allocation ($p = 0.056$–0.715, 5th–95th percentiles); at midpoint ages its p is 0.0085, and with every age at a shared quantile of its range it spans 3×10^{-9} to 0.0165 (Supplement Table S7).

A history that fails under the allocations drawn may still fit another that the ranges admit. We therefore searched all admissible allocations exhaustively for the one most favourable to each history, including whether each of the two units that straddle the Emsian cutoff falls inside the window (Supplement S5.2). For crown-member bodies, no admissible allocation lets the 10 Myr decline reach nominal $p = 0.05$. The 25 Myr decline reaches it when one or both straddling units are counted (p up to 0.077), but not with the 48 guaranteed units alone ($p \leq 0.041$). Plant-excluded collection density combined with the 10 Myr decline also has a passing allocation. For land-plant bodies with the Oman unit, none of the six histories has one. The search chooses ages to favour each history and treats the recorded ranges as independent, so its p values are not calibrated: it establishes what the recorded ranges permit for this diagnostic, not which history is true. The steep decline fails the diagnostic under every admissible age assignment; the 25 Myr decline does not; constant intensity is not shown to be adequate.

The sequence says nothing about the rate before the oldest possible crown-member find, so we asked how

much rarer earlier records would have to be for a recoverable record to begin at a given older age. With a step in the rate at the oldest possible find (426.7 Ma) and no assumed delay before diagnostic bodies, a start at 431.2 Ma needs earlier records at 0.48 of the later rate, and starts at 444, 447 and 452 Ma need 0.124, 0.105 and 0.085 of it (Supplement Table S9b). Under the coarser region–stage or curated units, 431.2 Ma needs no reduction at all and 444–452 Ma needs 0.22 to 0.42 of the later rate. These are retrospective compatibility contours: the step sits at an age chosen after seeing the data, the 0.05 rule gives them no nominal confidence, and they are not fitted rates. The tenfold scenario in Table 3 is one point near this frontier. The fitted records look the same under every point on it, which is where the answer is decided: in an interval the body record cannot see.

Finally, a recoverability endpoint is not a crown date. Mapping one to the other needs an assumed delay between crown origin and the recoverable presence of crown-diagnostic bodies (Marshall 2019). With constant intensity, reaching the young end of Harris et al.'s interval (447 Ma) needs an assumed delay of at least about 18 Myr at the 95% endpoint bound; a 20 Myr delay maps 428.8 Ma to 448.8 Ma. These are minimum offsets, not fitted delays, and the record does not estimate the delay. Delay and earlier recording rate trade off against each other; Supplement Table S9b gives the frontier under assumed delays of 5 to 20 Myr.

Simulations at the real unit counts confirm that the calculation behaves as stated when its assumptions hold (exact-age coverage 0.941–0.958) and show how it fails when they do not. Duplicated horizons treated as independent cut coverage to about 0.78 (pairs) or 0.53 (quartets). Fitting constant intensity to a true 10 Myr decline gives 0.03–0.24. An ignored 10–20 Myr origin-to-body delay can give zero coverage of the lineage age (Supplement S5.6, Table S11).

4. What follows for early land-plant evolution

4.1 An older, sparsely recorded flora is allowed, not required

Taken together, the three analyses support a specific, conditional conclusion. Where the record can be interrogated, it rules out early land plants that were both abundant and readily recorded before the younger dates: no well-delivered, Llandovery-level trilete producers sustained through the later Ordovician in the Murzuq Basin, and no crown vascular-plant bodies recorded at the later Silurian–Devonian rate before about 429 Ma, under the primary unit rule. It does not rule out early land plants that were rare, local, small, poorly preserved or seldom recognised. Murzuq allows trilete producers at a tenth of their later intensity, at full intensity with poor delivery, or living outside the basin's catchment. The body record allows crown vascular plants that were present but recorded at a tenth of later rates, or that lacked crown-diagnostic bodies, before the Ludlow.

Whether those allowances are plausible is a biological question that the record cannot settle directly. Larger size, cuticle and lignified tissue raise the chance that a body is preserved, found and recognised, and those traits evolved along the lineages being dated; yet the Ordovician Oman fragments show that small sporangia without such tissues can survive. Terrestrial deposits are scarce before the latest Silurian (Kenrick et al. 2012), and the oldest branched land-plant sporophyte is preserved in marine sediments (Libertín et al. 2018). In our judgement, a tenfold lower rate of recorded, identifiable crown-member occurrences before the Ludlow is not extreme, but neither is it demonstrated. Including or excluding a single fragment assemblage moves the land-plant body bound by about 20 Myr, which shows how strongly such bounds depend on their oldest admitted records; it does not measure how long land plants lived before their bodies became recoverable.

What this means for the clocks is an audit of assumptions, not a ranking. Under constant or effort-tracking intensity and the primary unit rule, the crown vascular-plant bound lies inside Hooper et al.'s 432–428 Ma interval and inside the full Morris et al. range (451–419 Ma). The Morris et al. means near 444 Ma and Harris et al.'s interval need either a long early phase in which crown vascular plants were recorded as identifiable bodies about 8 to 12 times less often than later (under the primary unit rule; 2.4 to 4.5 times under coarser ones), or crown-diagnostic body features that arose at least about 15–23 Myr after the crown node (minimum offsets at the constant-intensity bound), or some combination. Constant intensity is algebraically simple, but nothing shows it is biologically appropriate near a clade's origin, so agreement between its bound and the youngest clock is not independent support for that clock; the same fossils are among its calibrations. Soft-bound maxima are a legitimate way to express this uncertainty. What the record asks of them is that the

mass and shape of the tail be justified against an explicit recording history and character history, rather than chosen by convention.

4.2 Which conclusions depend on which assumptions

Table 4. Dependence of each main conclusion on recording assumptions and on the character-to-lineage mapping.

Conclusion	Rests on	Fails if
Murzuq excludes abundant, well-delivered trilete producers sustained through the later Ordovician	encounter model; delivery ρ; reference purity; stage dating of cuttings, chiefly 18 later-Ordovician ones; independence of cuttings within a well	older delivery was much lower, reference triletes include caved or recycled grains, ages are shifted, or cuttings from one well and formation behave as one observation (limit about 0.4)
That exclusion bears on crown land plants	producers lived in the catchment; trilete production at the embryophyte crown and its persistence (Hooper et al. 2026); our persistence pilot	early producers lived elsewhere, did not make recognisable trilete monads, or lost them in lineages present in the basin
Recoverable crown vascular-plant bodies began after about 429 Ma	recording history before and within the record; unit rule and independence; ages within reported ranges; adopted stage boundaries	earlier recording was several times lower (§3.3); units are pseudoreplicated or coarser (region–stage or curated units give 432.4–434.05 Ma); Yea or Bathurst ages are wrong
That bound bears on the crown vascular-plant node	lycophyte-line and euphyllophyte bodies are crown members; crown-diagnostic bodies arose near the node	crown origin preceded recognisable crown bodies by a long delay
The land-plant body bound is 455.2 Ma under constant intensity	the Oman fragments are one unit of the declared category; recording history	the fragments are excluded (434.5 Ma) or split; the poorly fitting histories misdescribe the sequence

The right-hand columns of Table 4 separate assumptions about how fossils were recorded from assumptions about what the fossils were. More searching of empty rock cannot settle the second kind: no number of barren samples shows that a body with sporangia but no demonstrated tracheids belongs to the vascular crown, or that a trilete mark means an embryophyte producer. Better-preserved specimens can, by adding characters. Our pilot of specimen-to-node links shows that some links are supported under declared models. For example, under stochastic-mapping models fitted to Hooper et al.'s deposited data, a trilete aperture present at the land-plant crown persists to the vascular-plant crown with probability 0.977. That figure is conditional on the character coding, model, tree and chronology; it is not the share of extinct local producers that made recognisable trilete spores. Other links, notably from the oldest branched sporophytes to vascular crown membership, remain contested (Supplement S6).

4.3 What would sharpen the answer

Four kinds of evidence would most improve these inferences. Any author could use them, and none is requested here.

1. Documented searched-but-empty Silurian non-marine beds with stated effort, which would turn the body record's unobservable early recording rate into an estimable quantity.
2. A second older-empty, younger-positive spore sequence outside polar Gondwana, with counted target and control grains, and an enumerated pre-515.5 Ma marine sequence with land-plant and algal counts.
3. Finer, independent dating of the Yea and Bathurst Island horizons, which set the crown-member bound.

4. Characters placing the oldest polysporangiate bodies on the lycophyte or euphyllophyte side of the vascular crown; tracheids alone would improve their placement but would not settle it.

5. Conclusions

The fossil maxima that most divide land-plant clocks cannot be justified or rejected by absence alone. The oldest combines disputed affinity with a qualitative preservation-control argument that the assembled material cannot quantify. The second has one qualified local test, in one polar basin, which excludes abundant, well-delivered producers sustained through the later Ordovician there, but not sparse, poorly delivered or absent-from-the-catchment ones, and whose strength depends on treating cuttings as independent. The third is consistent with the positive body record under constant or effort-tracking recording and the primary unit rule, and becomes compatible with the older clocks if identifiable crown vascular-plant bodies were recorded several times less often before the oldest find than after it, by a factor that depends on how the records are counted. Bounds of this kind are set by their oldest admitted records: one Ordovician fragment assemblage moves the land-plant body bound by about 20 Myr. An older, sparsely recorded early flora is therefore a live possibility under assumptions that the record itself cannot test. Clock analyses should treat these maxima as conditional on stated recording and character assumptions, and propagate them as such.

Methods in brief

All ages use the ICS v2024/12 stage bases held in the project timescale. Literature dated to GTS2012 was converted at stage level, with source-specific statements retained. Each analysis declares a named reading of disputed identifications (main or inclusive). An explicit non-detection is scoped to the material examined; a missing report becomes a zero only after a full reporting audit. The Murzuq encounter model, its simulation-based qualification and the contamination identification result are in Supplement S2. The QSIM floor calculation and its proof are in S3, and the Cambrian affinity review in S4. The calibration material at Borenshult-1 is in S2.4. The body-record inventory, unit rules, age envelope, recording histories, shape diagnostics, the Oman fragments and their exclusion sensitivity, sensitivity grids and simulations are in S5, and the specimen-to-node pilot in S6. Supplement S7 records analyses superseded during review and why.

Roles of AI systems and humans

Barry W. Brook set the programme and its stages and directed the reconstruction of this paper. He authorised the AI collaborators to proceed on routine decisions and retains decisions on external correspondence, release and submission. This is an internal draft awaiting his final review.

Claude Opus 5.5 (Anthropic, `claude-opus-5-5`, in Claude Code) developed the absence-test framing and engine and led the first modelling loop. It proposed the QSIM floor argument, reviewed the Cambrian affinity sources, and proposed and piloted the positive body-record analysis. It also drew the manuscript figures and wrote this reconstruction and its supplement. It adjudicated the second agent's manuscript review and added the Oman fragment sensitivity.

OpenAI Codex (GPT-6; the exact variant was not exposed to it) implemented the Murzuq observation audit and the second loop's corrections. It reviewed the body-record pilot and replaced its estimator, units and source matching after finding material defects, then implemented the reviewed body-record analysis, simulations and sensitivity grids. It reviewed the reconstructed manuscript, rechecked the Borenshult trilete readings and screened them, and compared the earliest trilete records on common criteria. In response to the GPT-6 Pro review it rebuilt the body inventory with an executable membership test, finding and repairing the polysporangiophyte admission error, and implemented the age–intensity frontier, interval-feasibility search, shared-timescale scenarios, Murzuq dependence sensitivities, influential-record checks, value-bound claim checking and the public raw-input rebuild. Claude triaged that review, corrected the account of the 515.5 Ma maximum, recomputed the new analyses' key values independently and integrated them here.

Bounded delegates of both systems contributed. `gpt-6.1-sol` with web search classified the 241 PBDB plant names with cited sources. Codex and Claude subagents independently checked the mathematics, the transcribed tables and the source claims. Each agent reviewed the other's work. Agreement between the two is not independent validation, and disagreements are kept visible in the audit record. Barry W. Brook

commissioned an independent review of the repository at commit `60c406b` from GPT-6 Pro (OpenAI). It reproduced the central calculations and identified a misstated calibration rationale (M1), the Oman inventory consequence, the composite nature of the recording rate, the temporal support of the Murzuq zeros and limits of the shape diagnostics; this revision acts on it (`docs/audits/2026-11-07-gpt6pro-triage.md`). Exact roles, versions and reviews are in `docs/charter.md`, `docs/decisions.md` and `docs/audits/`.

Data and code

The repository holds the source-linked derived tables, model code, frozen qualification evidence, reviewed body-record outputs with hash receipts, figure and table scripts, and the audit trail. Copyrighted source texts and third-party deposits are not redistributed; scripts fetch the deposits and verify their hashes. The frozen PBDB all-taxa collections extract that the effort histories use is archived under CC0 at https://doi.org/10.5281/zenodo.24028289, so the derived tables can be rebuilt from raw input (Supplement S8). Public release of the paper is Barry W. Brook's decision.

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Fig. 1.

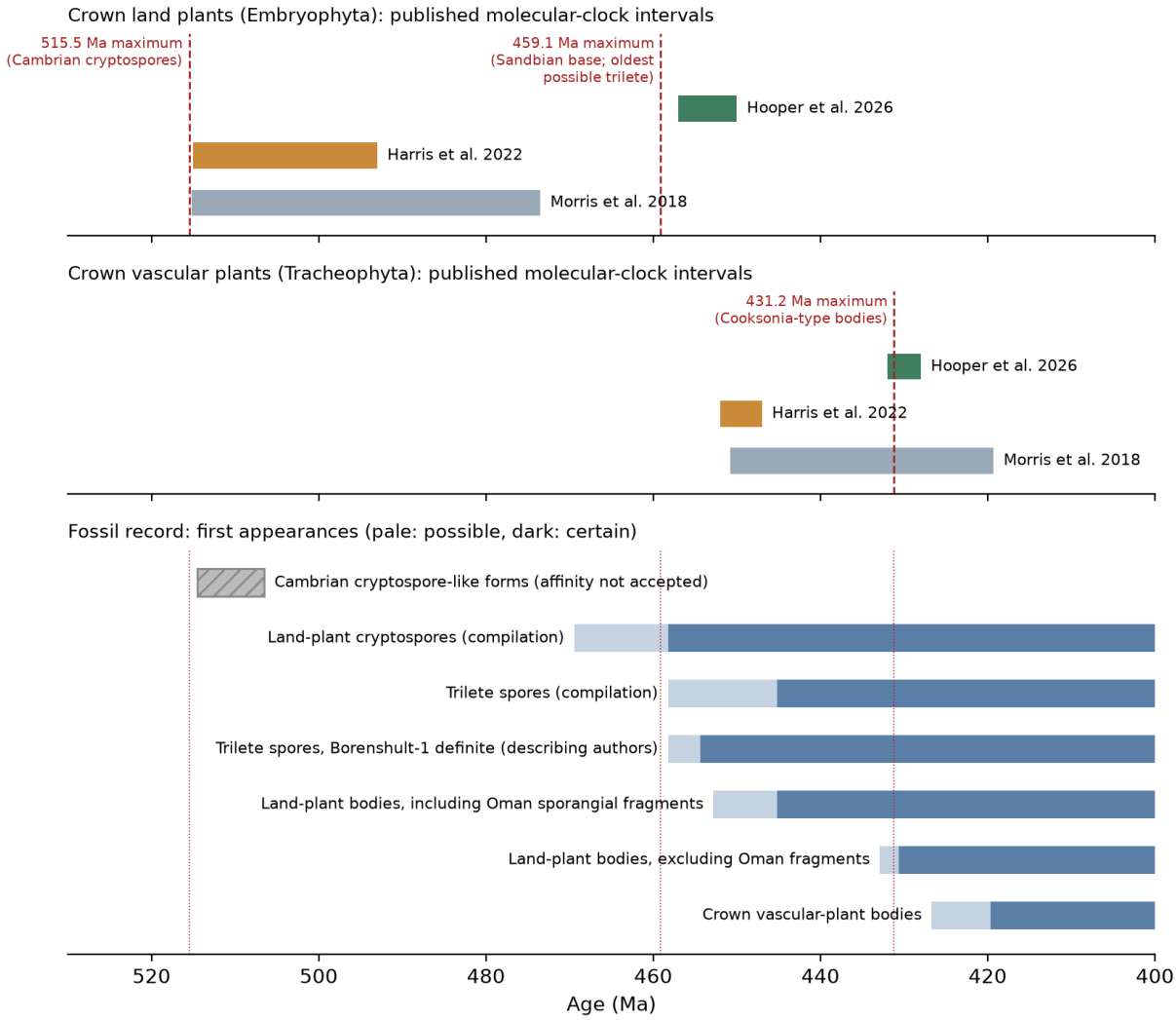


Figure 1: The problem. Top two panels: published crown-age intervals from three molecular-clock studies (Morris et al. 2018b; Harris et al. 2022; Hooper et al. 2026), with the three calibration maxima examined here (dashed). The bars do not share one statistical meaning: Morris et al.'s are envelopes of 95% highest-posterior-density intervals across seven topologies; the others are the intervals each study reports. Bottom: first appearances by source and reading. Pale bars run from the oldest possible to the oldest certain occurrence. "Compilation" bars are the strict-set first appearances of the project spore compilation. The Borenshult-1 bar is the describing authors' definite-trilete reading (Rubinstein and Vajda 2019): Sandbian, below the 454.41 Ma bentonite; it is shown beside the compilation, not substituted for it. Body bars are the reviewed locality units of this study: land-plant bodies including the Katian Oman sporangial fragments (canonical) (Wellman, Osterloff, and Mohiuddin 2003) and, for comparison, excluding them. Dark bars mark the span after the oldest certain occurrence, not a continuous record. The 459.1 Ma maximum is the GTS2012 Sandbian base plus its uncertainty. The hatched bar marks Cambrian Stage 4 cryptospore-like forms, whose land-plant affinity none of the sources examined accepts.

Fig. 2.

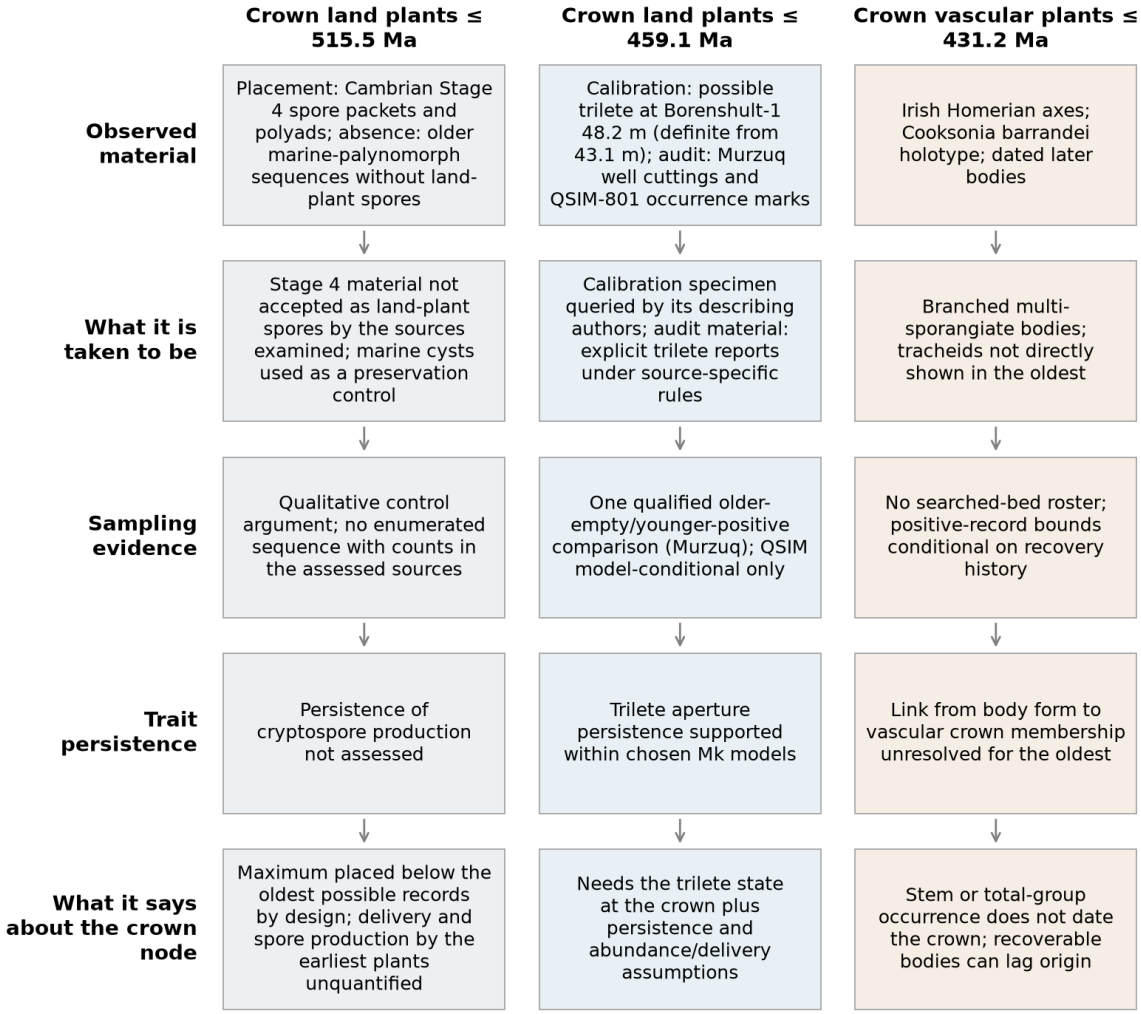


Figure 2: From observed material to node interpretation. For each maximum, the observed material, what it is taken to be, the sampling evidence, the persistence of the diagnostic trait and the implication for the crown node are separate links; a maximum needs all five.

Fig. 3.

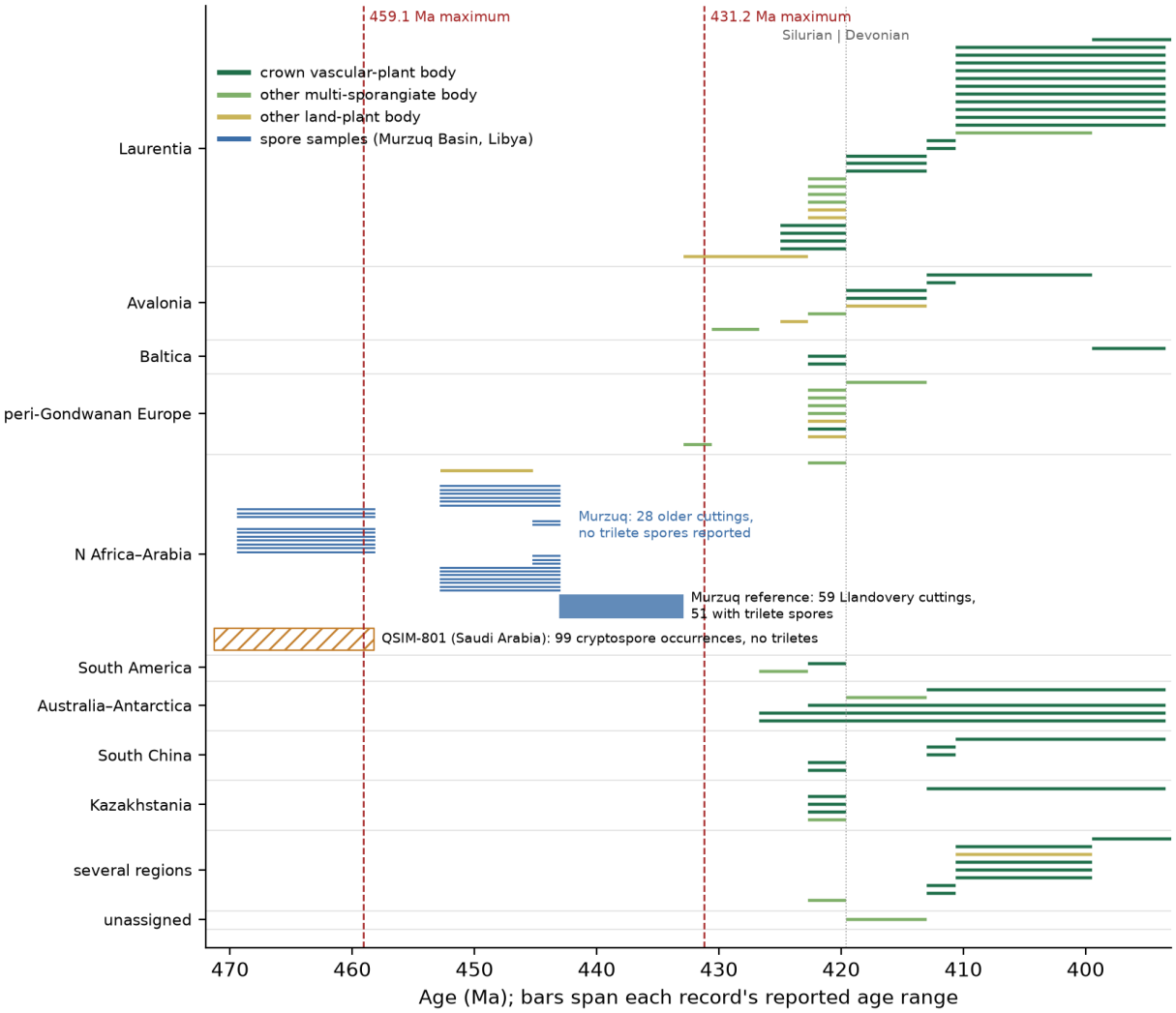


Figure 3: Where the tested evidence comes from. Body-fossil locality units (main reading, canonical inventory including the Oman fragments) by region; bars span reported age ranges, coloured by the most specific category each unit supports. Blue: Murzuq Basin cuttings used in the spore test (older cuttings without trilete reports, and the Llandovery reference block). Hatched: QSIM-801 cryptospore samples (full-stage enclosure; the source assigns them more finely, to the late Dapingian–early Darriwilian). Regions are approximate modern country or state assignments, not a palaeogeographic reconstruction. Spore records outside these two frames are not shown.

Fig. 4.

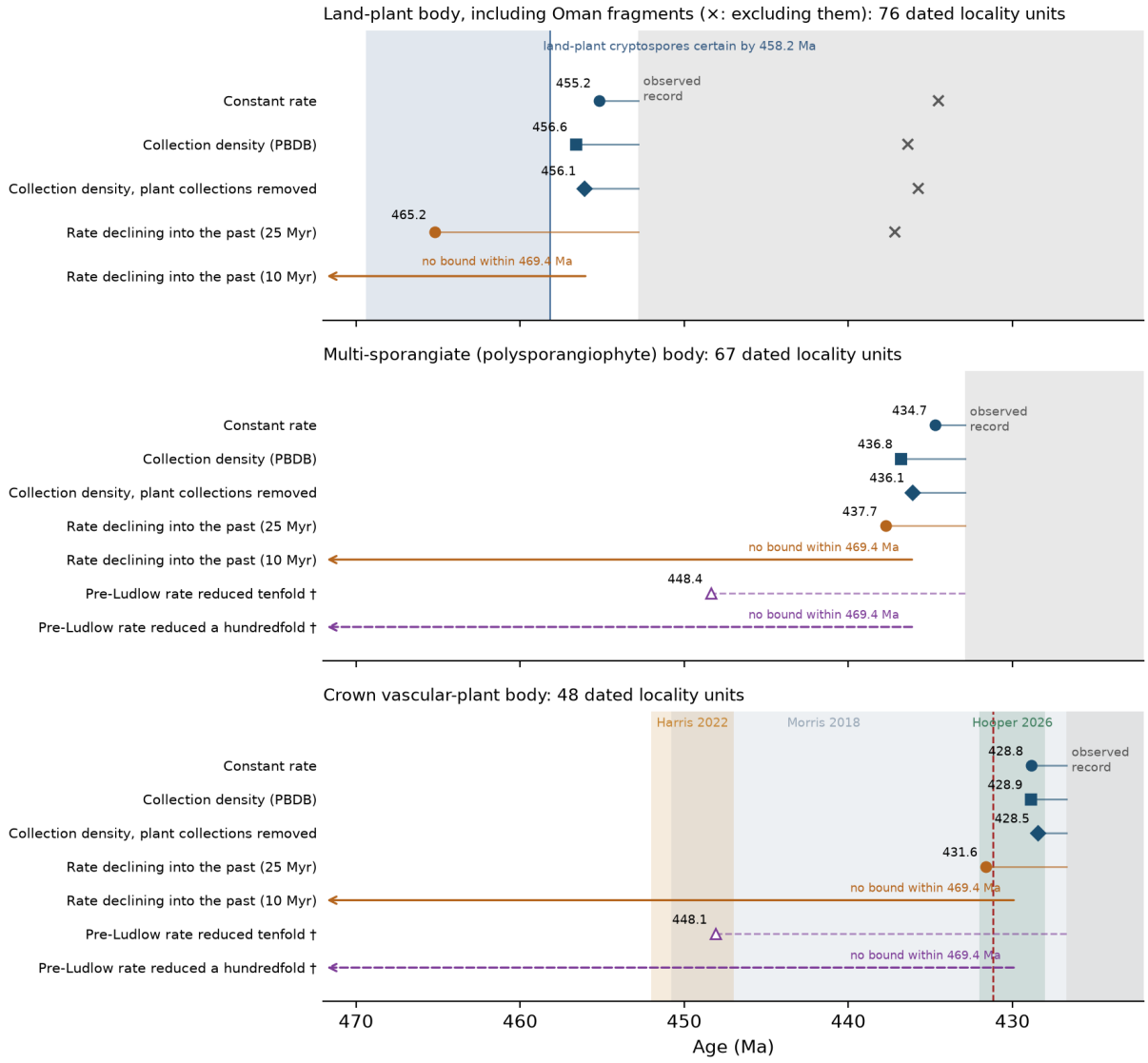


Figure 4: Older bounds on the start of the recoverable body record. Each row is one declared history of the rate of recorded, identifiable occurrences; points are conservative 95% older bounds conditional on that history, the adopted timescale and independent locality–age units (main reading). Grey: span of the observed units. Top: land-plant bodies including the Katian Oman sporangial fragments; × marks the bounds when they are excluded. These bounds are sensitivity arithmetic, not dates at which bodies became recoverable. Open triangles (†) are illustrative scenarios, chosen after the analysis, in which the rate before 426.7 Ma is reduced; for the crown-member category they leave the fit to the observed record unchanged. Top: blue shading marks land-plant cryptospores, possible from 469.4 Ma and certain by 458.2 Ma. Bottom: shaded bands are published crown vascular-plant intervals and the dashed line is the 431.2 Ma maximum; they are comparators, not validation targets. Arrows: no bound within the 469.4 Ma limit of the declared histories.